3400 · NORTH END

CIVIC DATA INFRASTRUCTURE · PROPERTY INTELLIGENCE · EQUITABLE REINVESTMENT

PRICING THE UNPRICED

3200 · CORE SPINE

A reproducible data engine that reads an address three ways — what happened here, what the record and market show now, what could change — piloted on the Greenmount/Waverly corridor and designed for eventual Baltimore-wide deployment.

3000 · PEABODY

100,867	~354	19	13–14%
NATIONAL REGISTER LISTINGS (EXTERNAL BENCHMARK DATA)	CORRIDOR PROPERTIES · GROUND TRUTH	WAVERLY WHITE PAPERS · MHT GRANT	PRIOR-STUDY BENCHMARK · ATLANTA (PATRICK 2025)

2700 · ENP CORE

SPONSOR OF RECORD **BALTIMORE NEIGHBORHOOD BUSINESS DISTRICTS (JEFF JETTON, VP)** · STRATEGY **OXCART ASSEMBLY** · PRODUCTION **NICEWORKS**
FACULTY **MANNY PATOLE, NYU CUSP** · PRIMARY CUSP FOCUS AREA **URBAN INFRASTRUCTURE** · TWO SEMESTERS · SEPT 2026 – MAY 2027 · MINDHIVE · DUE JULY 24, 2026

2500 · SOUTH END

CONTENTS

PROJECT SUMMARY

01 THE VALUE THE RECORD CAN'T SEE

Property-data systems describe parcels, assessments, permits, vacancies, and available programs. They do not systematically assemble documented historical provenance, or test whether recognition of that history is reflected in value.

The market prices the dirt and the building; the history goes unpriced. And that gap is widening: value increasingly pools in what cannot be copied — the authenticated fact that something happened *here*, and the unpriced option on what happens next. This capstone builds the instrument that measures the first and bounds the second: a reproducible **provenance evidence layer** assembled from public and sponsor-provided records, validated against property-market outcomes, and connected to an existing address-based public tool.

The committed pilot geography is the **Greenmount/Waverly corridor**. Students build the entity-resolution pipeline, the temporal people–place–event knowledge graph, an auditable **Provenance Index**, the corridor validation, and a working address-based pilot. A selected external recognition analysis (National Register listings against national transaction data) may be undertaken if data access, parcel linkage, and team capacity satisfy pre-established feasibility thresholds. Students deliver the documented pipeline, model card, validation report, working pilot, and productionization specification; **BNBD and Niceworks are responsible for deployment, maintenance, and future citywide scaling**.

> PROGRAM VISION VS. CAPSTONE SCOPE
>
> **Pricing the Unpriced is the first phase of a larger Economics of Provenance research program. The capstone does not attempt to complete that program.** Its committed contribution is the shared measurement infrastructure: a reproducible provenance evidence layer, a temporal knowledge graph, an auditable index, a corridor validation result, and a working Greenmount/Waverly pilot. National causal analysis, future-value modeling, and creative public modules are conditional extensions or later phases built on that foundation.

01 THE GROUND, MEASURED

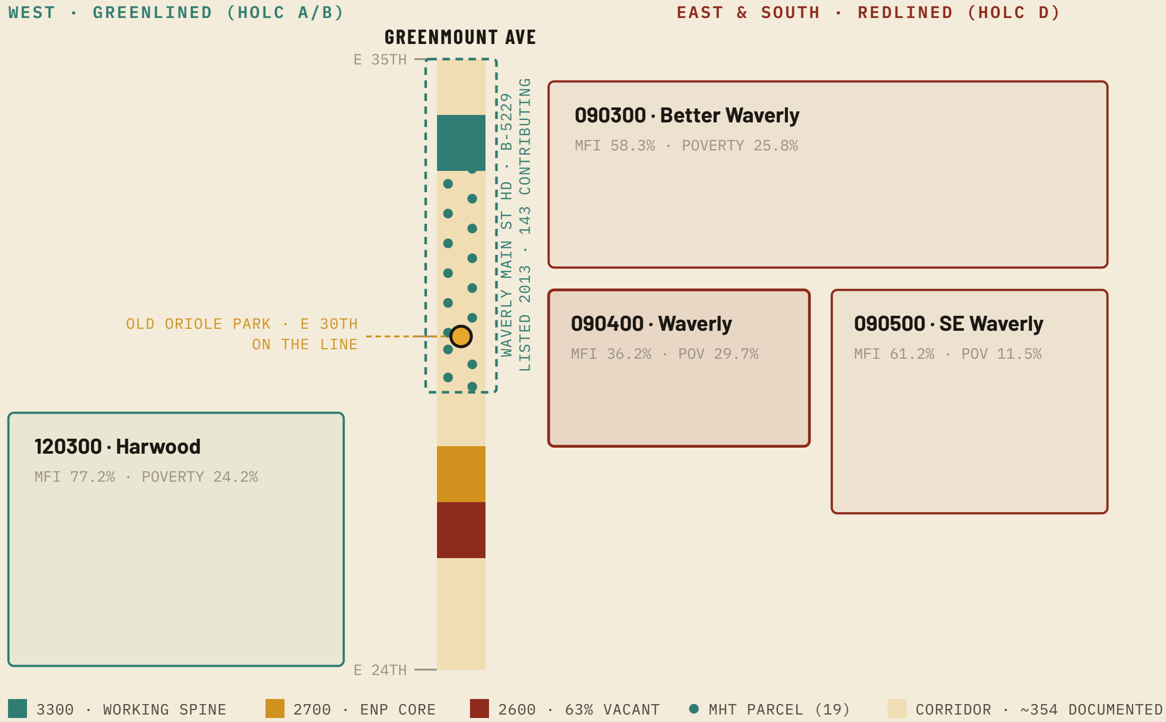


FIG. 1 — THE PILOT GEOGRAPHY, MEASURED. THE CORRIDOR SPINE — GREENMOUNT AVENUE, 2400–3399 (E 24TH–34TH) — WITH THE AXIS EXTENDED TO E 35TH WHERE THE WAVERLY MAIN STREET HISTORIC DISTRICT ENDS (THE DISTRICT BRACKETS E 29TH–35TH), THE 19 MHT WHITE-PAPER PARCELS, AND THE FOUR CENSUS TRACTS. TRACT GEOMETRY GENERALIZED FROM CENSUS TIGER/LINE; FIGURES PER ACS 2020–2024. BALTIMORE-WIDE DEPLOYMENT AND THE EXTERNAL VALIDATION SAMPLE SIT BEYOND THIS FRAME — DESIGN TARGET AND CONDITIONAL SAMPLE, NOT COMMITMENTS.

> **The division of labor is the whole design. Students build the engine — the evidence pipeline, the graph, the index, and the validation that decides what the public tool may claim — and a pilot wired into the sponsor's existing interface. BNBD / Niceworks productionizes and maintains the public tool; students are never the long-term maintainers. The graded artifact is the data science, not the app.**

One mile holds the proof and the need: the working northern spine shows what the corridor becomes; the 2600 block is why the evidence layer matters. A pilot that explains value across this gradient has faced the hardest local case there is.

Segment roles, occupancy, and vacancy per the sponsor's corridor market study and OZ 2.0 nomination case (sponsor-prepared). District: NRHP ref. 13001020, listed 12/31/2013. Tract geometry generalized from U.S. Census TIGER/Line 2020; demographics ACS 2020–2024. MHT parcel positions indicative within the studied blocks (2958–3360).

PROBLEM & MOTIVATION

02 THREE GAPS, ONE CORRIDOR

01 · THE MEASUREMENT GAP

HISTORY HAS NO COLUMN

A closely related recent study (Patrick 2025, Atlanta) finds National Register recognition associated with a 13–14% price premium while local regulatory designation trends toward null — markets respond to recognized history, yet no general, reproducible instrument measures the history itself. That premium is a benchmark from prior literature, not a result of this project.

02 · THE CIVIC GAP

A CITY-SCALE NEED

Baltimore is pursuing one of the nation's most ambitious vacant-property efforts (roughly 37,500 properties, a $5B private-investment target) and a reform agenda built on centralized, accessible information. No public tool yet lets residents see what a property is, what the record shows, and what they can do with it. The project is independent: not commissioned by, for, or delivered to the City. Conversations with city officials are ongoing; a formal expression of interest would follow the City's own review process.

03 · THE TOOLING GAP

RECORDS WITHOUT INTELLIGENCE

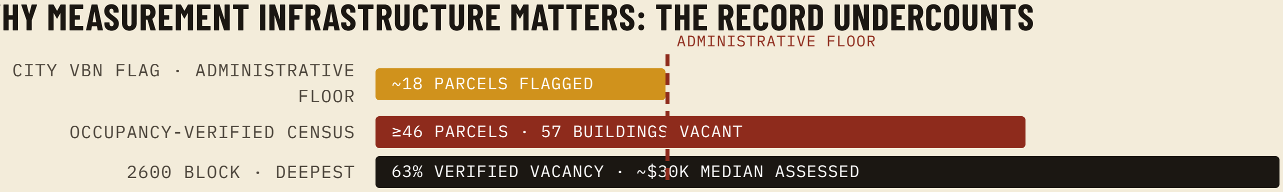
The sponsor's current tools are strong research aids but stop at "what's on record": no assembled history, no evidence score, no fusion of history with action — and no way to tell a resident what the documentation means.

THE CORRIDOR, ON THE RECORD

The pilot geography is not a blank slate — it is one of the best-documented commercial corridors in Baltimore, profiled parcel-by-parcel in the sponsor's Opportunity Zone 2.0 nomination case:

202	**154**	**1920**	**143**
COMMERCIAL PARCELS PROFILED (2400–3399)	DISTINCT OWNERS — THE FRAGMENTATION PROBLEM	MEDIAN BUILD YEAR — A HISTORIC FABRIC	CONTRIBUTING RESOURCES · WAVERLY MAIN ST HD (REF. 13001020)

WHY MEASUREMENT INFRASTRUCTURE MATTERS: THE RECORD UNDERCOUNTS



The occupancy census identified roughly 2.6× as many affected parcels — and more than 3× as many vacant buildings — as the administrative flag. When the record undercounts something as visible as vacancy, the case for rigorous measurement infrastructure — for history, evidence, and value — makes itself.

Parcel, owner, and build-year figures: City of Baltimore Real Property file, 2400–3399 Greenmount, via the sponsor's OZ 2.0 nomination case (July 2026). Vacancy: City VBN-Open vs. sponsor occupancy census, disclosed as sponsor-prepared. Historic district: Waverly Main Street HD, NRHP ref. 13001020, listed 12/31/2013 — 168 resources, 143 contributing.

RESEARCH QUESTIONS — TIERED BY COMMITMENT

RQ TWO COMMITTED QUESTIONS. THE REST EARN THEIR WAY IN.

COMMITTED **Primary question.** Can a reproducible provenance evidence layer and temporal narrative graph be constructed from public and curated historical records — and does the resulting index add meaningful explanatory or predictive information beyond conventional property and neighborhood data?

COMMITTED **Operational question.** Can the engine be responsibly operationalized in an address-based public pilot, with every output traceable to evidence and accompanied by appropriate uncertainty and limitations?

Why the conditional questions matter for equity. If documented history is associated with value, documentation is a form of capital — one historically disinvested, redlined neighborhoods were denied. A public instrument makes it possible to ask whether *equivalent* provenance shows a weaker association with value where devaluation was engineered, and to put that knowledge in residents' hands rather than only speculators'. This equity framing draws on Dr. Lawrence T. Brown's scholarship on Baltimore's racialized geography (*The Black Butterfly*) and on project discussions in which he has participated.

CONDITIONAL Does formal historic recognition capitalize into value (the external National Register analysis)? Does narrative connectivity generalize beyond the corridor? Does the association between documented provenance and observed value vary across historically redlined areas? Can completed physical improvements support a reliable improvement-scenario model?

LATER PHASES The separate causal effects of documentation, recognition, regulation, and investment; certification experiments; diffusion through narrative networks; activation-and-protection design that avoids displacement; whether public recognition can create reparative value without accelerating extraction.

Falsification. The committed core can lose: if the index adds no held-out explanatory power beyond conventional controls, or corridor connectivity adds nothing beyond physical proximity, the engine's monetary layer does not ship — the pilot then presents documented evidence without a value estimate. A public score that predicts nothing would not be displayed as one.

THE PRODUCT & THE ENGINE

03 PAST, PRESENT, FUTURE — KEPT HONEST

The product is an **address-driven property-intelligence lookup**, piloted on the Greenmount/Waverly corridor and designed for eventual Baltimore-wide deployment. It keeps three different kinds of information visibly distinct — observed facts, modeled evidence, and bounded scenarios — so a resident can trust each for what it is. The first two panels are committed; the third always shows applicable programs and evidence-backed actions, and displays a projected value scenario only when the conditional model clears its thresholds.

greenmountcorridor.com/lookup · 3200 Greenmount Ave, Baltimore, MD 21218

WHAT HAPPENED HERE

ILLUSTRATIVE COMPOSITE · DRAWN FROM DOCUMENTED CORRIDOR RECORDS

1912 · storefront built; first tenant a grocer	directory
1939 · block anchors a championship season	newspaper
1993 · trademark registered at this address	USPTO
2013 · listed, Waverly Main Street HD	NRHP

In production, each entry links to its source and evidence status for the queried address.

WHAT THE RECORD SHOWS NOW

OBSERVED

Assessed value (2026)	**$129,733** · SDAT
Land use	**Commercial (C)** · SDAT
Sales on record	**3** · liber/folio

PROVENANCE EVIDENCE · ILLUSTRATIVE

VERIFIABILITY	.90
DENSITY	.72
SALIENCE	.81
CONNECTIVITY	.66

A dollar interpretation appears only above the validation threshold.

WHAT COULD CHANGE

SCENARIO · ILLUSTRATIVE · STATED ASSUMPTIONS

Post-improvement value range	**$175K–$210K**
Improvement cost	**$65K**
Potential incentive coverage	**$32K**
Owner contribution	**$33K**

Figures are illustrative only and do not represent a completed valuation or confirmed program award. Never an appraisal, promise, or guaranteed return.

FIG. 2 — THE PILOT'S THREE-PART ADDRESS VIEW (ASSESSED VALUE AND LAND USE ARE THE REAL SDAT RETRIEVALS FOR 3200 GREENMOUNT; THE HISTORY PANEL IS AN ILLUSTRATIVE COMPOSITE OF DOCUMENTED CORRIDOR RECORD TYPES; SCORES AND SCENARIO FIGURES ARE ILLUSTRATIVE). OBSERVED, ESTIMATED, AND SCENARIO CONTENT CARRY DIFFERENT LABELS AND DUTIES.

WHAT ALREADY EXISTS (SPONSOR-PROVIDED — THE HEAD START)

- **Property Dossier (Niceworks).** Live Esri geocoding + Open Baltimore parcel layers + Maryland SDAT roll; cycles 22 curated sources across six categories; corridor-specific intelligence (the pre-1919 York Road renumbering); role-based oral-history capture; emits a cited PDF.
- **Improve Your Property.** Address → matched free BNBD programs — the action layer the scenario panel would draw on.
- **The curated corridor knowledge base.** ~354 documented properties, ~2,600 documented incidents, ~150 registered intellectual-property records (patents and trademarks), and the people–place–event relationships the graph formalizes.
- **The Waverly Main Street Study.** A funded Maryland Historical Trust FY2026 grant (18 months, May 2026 – Oct 2027, with an assigned MHT project monitor) producing per-property historical white papers across 19 parcels in district B-5229.

03 WHAT STUDENTS BUILD, WHAT WAITS ITS TURN

WHAT STUDENTS BUILD (THE COMMITTED ENGINE)

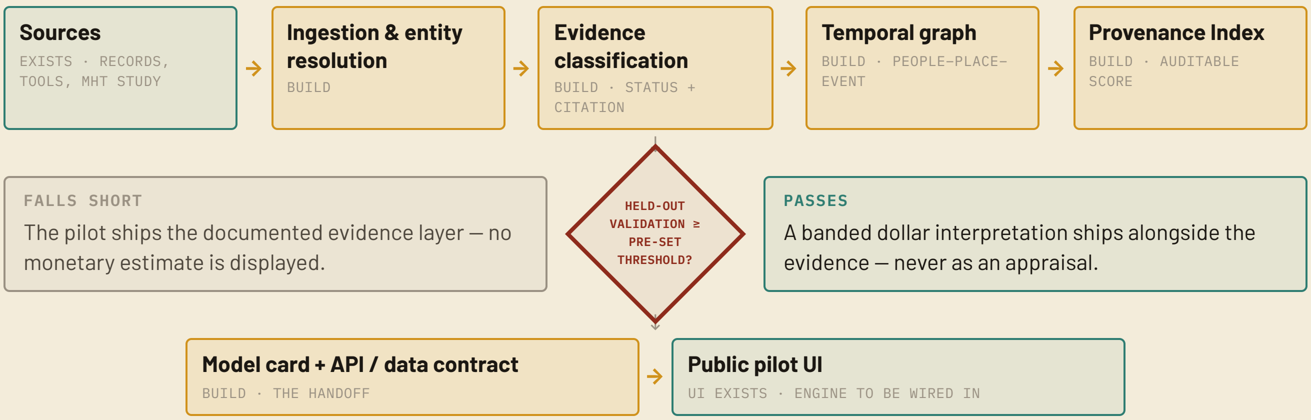


FIG. 3 — THE ENGINE AS A GATED PIPELINE. TEAL = ALREADY EXISTS (SPONSOR); AMBER = STUDENTS BUILD. THE OXBLOOD GATE IS THE PRE-COMMITTED DECISION RULE: NO THRESHOLD, NO DOLLARS.

- COMMITTED A reproducible **provenance evidence pipeline**: ingestion, entity resolution, evidence-status classification, temporal processing.
- COMMITTED The **temporal knowledge graph and Provenance Index** — auditable relationships and a transparent parcel-level evidence score.
- COMMITTED The **corridor validation**: held-out testing, the connectivity-versus-proximity analysis, and the decision of whether a monetary interpretation may ship.
- COMMITTED A **working corridor pilot** wired into the existing interface, plus the model card, API/data contract, and productionization spec BNBD needs to deploy and maintain it.

CONDITIONAL EXTENSIONS (CAPACITY- AND FEASIBILITY-GATED; AT MOST ONE PROTOTYPE MODULE)

- CONDITIONAL The external National Register recognition analysis (§05); equity heterogeneity across redlined geography; the improvement-scenario model.
- CONDITIONAL An expanded program & incentive knowledge base; automated source connectors from the sponsor's *Maryland & Baltimore Property Research Index*; an open-data memorandum identifying the public datasets whose release would most improve property intelligence.
- CONDITIONAL One creative prototype, chosen by the team — e.g., generated façade-improvement sketches or automated, fully-cited property narratives.

> **Tiered commitment keeps this honest: the engine, validation, and pilot are committed. The conditional extensions listed above proceed only if the core remains on track, and are cut first if it slips.**

DATA SOURCES & ACCESS READINESS

04 WHAT'S IN HAND, WHAT NEEDS A KEY

Every source was checked for existence, content, and access path before this proposal was written. (Note: the NPS **Developer** API at developer.nps.gov serves park content, **not** National Register listings; the authoritative NRHP bulk data is the NPS Data Downloads + IRMA store.)

Fallback, stated up front. If ZTRAX access is delayed or its coverage is insufficient, the capstone completes the corridor pilot on SDAT, Open Baltimore, and selected assessor or transaction datasets, and narrows the external-validation geography accordingly. The committed core never depends on restricted data.

SOURCE	ROLE	ACCESS STATUS	NOTES
Corridor knowledge base & relationships	Ground truth + index inputs	**Available to sponsor now**	~354 properties / ~2,600 incidents / ~150 registered intellectual-property records; the labeled set the engine learns on.
Waverly Main Street Study white papers (MHT FY2026 grant)	Deep per-parcel documentation, 19 parcels (B-5229)	**In progress**	Funded 18-month study (May 2026 – Oct 2027) with assigned MHT monitor; papers land during the capstone.
Open Baltimore + Maryland SDAT	Parcels, assessments, sales, permits, vacancy — the corridor outcome data	**Publicly downloadable**	Already wired into the sponsor's live tools. Waverly B-5229 listed 2013-12-31.
national-register-listed.xlsx (NPS) + IRMA GIS	External benchmark treatment: 100,867 dated listings	**Publicly downloadable**	Verified by inspection (refreshed 2026-05-22); geometry via IRMA Ref. 2210280.
Zillow ZTRAX via ICPSR (Study 39652)	External-validation outcomes: national transactions + assessments, 1940–2020	**Restricted — application required**	Depends on NYU/ICPSR access and CUSP's secure Research Computing Facility. Not assumed.
HOLC polygons — Mapping Inequality	Redlining-exposure measure for the conditional equity analysis	**Publicly downloadable**	Paired with ACS and HMDA lending; historical exposure kept distinct from present-day conditions.
BNBD program / incentive data	The action layer; scenario inputs	**Available to sponsor now**	Already powers the live Improve tool.
federal-DOEs.xlsx (Determinations of Eligibility)	Considered as an "eligible-but-not-listed" control	**Rejected as unsuitable**	Rows are federal projects, undated, federal-proximity-selected — not a clean control.

METHODS — BUILD, VALIDATE, THEN INTERPRET

05 SCORE FIRST, DOLLARS LAST

STAGE 1 — A PRE-SPECIFIED PROVENANCE EVIDENCE SCORE COMMITTED

Construct a per-parcel score from **Verifiability** (evidence strength — a multiplier, so unverified claims are capped, not rewarded), **Density**, **Salience**, **Distinctiveness**, and **Connectivity** (graph centrality) — *without reference to sale outcomes*. The coding protocol is frozen and version-controlled early; coders are blinded to prices; inter-rater reliability is measured. The β and γ parameters are fixed in the coding protocol before any outcome data are examined; alternative values are evaluated only through pre-specified sensitivity tests and are never selected to maximize price prediction. Every score is auditable to its cited claims, and carries a source-coverage measure so that archival abundance is never mistaken for significance — archival silence is flagged, not scored as an absence of history.

$$\Pi(s) = \mathit{Verifiability} \times (\mathit{Density} + \beta \cdot \mathit{Connectivity}) \times \mathit{Salience}^{\gamma} \times \mathit{Distinctiveness}$$

EQ. 1 — THE INDEX (PRE-SPECIFIED)

STAGE 2 — INDEPENDENT VALIDATION COMMITTED

Test whether the score adds explanatory or predictive power beyond building characteristics, parcel characteristics, location, time, and neighborhood conditions — using spatially blocked (not random) and temporal holdouts, cross-validation, outcome-blinded coding, robustness testing, and **pre-established minimum performance thresholds**. The data-readiness audit designates the corridor's primary outcome before feature construction: verified transaction price where coverage is sufficient; assessed value as a separate administrative outcome, never represented as a market price. Spatial blocking with neighborhood and time controls limits geographic leakage between training and test parcels. Before model fitting, the data-readiness audit also establishes minimum usable-sample, transaction-coverage, temporal-coverage, and missingness thresholds; if these are not met, the monetary validation narrows to an adequately observed subset or remains exploratory. Performance is evaluated as incremental held-out explanatory power and error reduction relative to a conventional property-value baseline — out-of-sample R^2, mean absolute error, and calibration. The corridor connectivity test belongs here: among physically comparable parcels, does graph connection carry information proximity does not?

$$\ln(\text{value}) = f(\text{structure, parcel, location, time, neighborhood}) + \theta \cdot \Pi + \varepsilon \text{ — evaluated on held-out data only}$$

EQ. 2 — HELD-OUT VALIDATION

Temporal validity. For historical valuation tests, provenance claims, source records, and graph relationships are time-stamped and reconstructed as of the relevant transaction or assessment date. Information first documented after an outcome date is never used to predict that earlier outcome — a 2026 white paper cannot be treated as known to the market in 2005.

05 THE DECISION RULE, AND THE EXTERNAL TEST

STAGE 3 — MONETARY INTERPRETATION COMMITTED DECISION RULE

> **A dollar interpretation is calculated and displayed only if the index meets its held-out validation threshold. If it does not, the pilot ships the documented evidence layer without a monetary provenance estimate. That is a sign of rigor, not failure.**

CONDITIONAL EXTERNAL TEST — DOES RECOGNITION CAPITALIZE? CONDITIONAL

The National Register analysis is an **external benchmark for whether formal recognition capitalizes into market value** — it does not itself validate the Provenance Index, which is evaluated separately through held-out performance. If access and linkage are confirmed (go/no-go in early spring), the design is a staggered-adoption difference-in-differences (Callaway & Sant'Anna 2021; Sun & Abraham 2021): unit = listed property (or district-by-parcel with contributing status); treatment date = listing date; controls = not-yet-listed properties; primary outcome = log transaction price; sample = selected states/markets scaled to access; flags for local-regulation overlap and rehabilitation-tax-credit exposure; anticipation handled by testing parallel trends in the far-pre window (values are documented to move up to ~3 years pre-listing; Patrick 2025) with a nomination-date robustness anchor. Benchmark: Patrick's Atlanta estimate of 13–14%. Fallback per §04.

CONDITIONAL IMPROVEMENT-SCENARIO MODEL CONDITIONAL

Candidate improvement events are identified from closed permits, final inspections, certificates of occupancy, or other completion indicators, subject to a data-quality audit — an issued permit is not treated as completed construction. The scenario output keeps its components separate: projected post-improvement value range, improvement-cost range, potential incentive coverage, estimated owner contribution, and the resulting net-equity scenario. Costs are never subtracted from asset value and relabeled as "post-improvement value." Displayed only as a labeled scenario under stated assumptions.

METHODS — CONTINUED

05 THE EQUITY LENS, ON THE 1937 LINE

CONDITIONAL EQUITY HETEROGENEITY — INFORMED BY BROWN'S BLACK BUTTERFLY FRAMEWORK CONDITIONAL

GREENMOUNT AVENUE — THE 1937 LINE

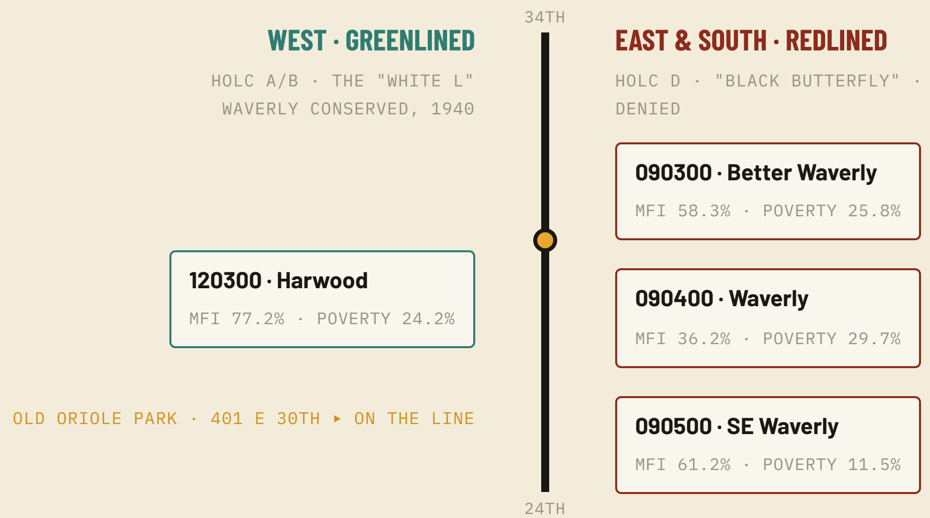


FIG. 4 — THE FAULT LINE. SCHEMATIC; HOLC 1937 GRADES GENERALIZED. TRACT FIGURES: ACS 2020–2024 VS. METRO MEDIAN FAMILY INCOME ($126,229). THE AMBER NODE IS OLD ORIOLE PARK, 401 E 30TH — DIRECTLY ON THE LINE.

One address makes the question concrete: the corridor's signature cultural asset — the lost Old Oriole Park diamond at 401 E 30th — sits directly on the 1937 line. Provenance, engineered devaluation, and reinvestment converge at one documented address; measuring that convergence is what the graph and this analysis are built to do.

$$\ln(V) = \alpha + \beta\cdot\Pi + \gamma\cdot X + \delta\cdot R + \lambda\cdot(\Pi \times R) + \varepsilon$$

EQ. 3 — HETEROGENEITY (CONDITIONAL)

R is a parcel- or block-group-level redlining-exposure measure built from HOLC polygons, kept distinct from present-day neighborhood conditions (which enter as controls); no single HOLC grade is assigned to the corridor as a whole. A negative interaction would be **consistent with a weaker association** between documented provenance and observed value in historically redlined areas; results are interpreted as heterogeneous associations, not causal effects, unless the identification assumptions support a stronger claim. Displacement and ownership-stability indicators are included only where reliable data exist. Equity analyses draw on Brown's published framework and on project discussions; this proposal does not represent his participation as formal review or endorsement.

Fault-line framing per Lawrence T. Brown, The Black Butterfly (2021); HOLC 1937 map via Mapping Inequality (Univ. of Richmond). Tract demographics: ACS 2020–2024 (B19113, B17001). This subsection recounts documented public history; no forward-looking causal claim is made.

STUDENT SKILLS EXERCISED

06 THE SKILLS, AND THE TWO SEMESTERS

SKILL AREA		HOW IT IS EXERCISED
NLP & information extraction	CORE	Entity extraction and evidence-status classification over deeds, newspapers, directories, and nomination documents; the assembled-history layer.
Graph construction & analytics	CORE	The temporal people–place–event graph; centrality for the Connectivity term; the connectivity-versus-proximity test.
Geospatial record linkage & panel construction	CORE	Knowledge base ↔ parcels ↔ SDAT/Open Baltimore (↔ NRHP/IRMA ↔ ZTRAX if the external test proceeds) in one reproducible pipeline.
Statistical validation & econometrics	CORE	Core held-out validation and robustness testing; causal inference (staggered DiD) if the external analysis clears go/no-go.
Applied ML & product data engineering	CORE	Hedonic baselines; the API/data contract and model card that let the engine feed a public tool responsibly.
Civic data / generative prototyping	CONDITIONAL	Source connectors, program knowledge base, open-data specifications, or one creative prototype (e.g., façade sketches, cited narratives).

FALL — BUILD AND DEMONSTRATE

WEEKS	ACTIVITIES	MILESTONE
1–3	Confirm data access; audit sponsor datasets; freeze evidence taxonomy and coding protocol; define temporal-validity rules and model/API contracts.	Data-readiness audit + frozen methodology spec
4–8	Normalize the corridor knowledge base; build entity resolution; create the temporal graph; compute Provenance Evidence Score v1; construct the corridor analytical panel.	Provenance Engine v1
9–14	Corridor-level validation; connectivity vs. proximity; wire a vertical slice into the existing Property Dossier interface; demonstrate selected addresses end to end.	Working corridor pilot + preliminary validation

WORK PLAN — CONTINUED

06 SPRING: VALIDATE, HARDEN, HAND OFF

FALL 1–3	FALL 4–8	FALL 9–14	SPRING 1–5	SPRING 6–10	SPRING 11–14
FREEZE & AUDIT	**ENGINE V1**	**CORRIDOR PILOT**	**VALIDATE & DECIDE**	**SELECTED VALIDATION**	**HAND OFF**
◆ FROZEN METHODOLOGY SPEC	◆ PROVENANCE ENGINE V1	◆ WORKING PILOT + PRELIM VALIDATION	◆ GO / NO-GO ON EXTERNAL DATA	◆ FINAL ANALYTICAL FINDINGS	◆ ENGINE · PILOT · DOCS · FINAL CUSP SHOWCASE

SPRING — VALIDATE, HARDEN, HAND OFF

WEEKS	ACTIVITIES	MILESTONE
1–5	Held-out and robustness testing; improve entity resolution and evidence classification; document uncertainty and failure thresholds; external-analysis go/no-go decision.	Validated corridor engine + external decision
6–10	One selected external validation analysis if feasible — otherwise deepen corridor/selected-market validation; equity heterogeneity if the data support it; at most one conditional prototype module.	Final analytical findings
11–14	Finalize pilot, model card, API/data contract, reproducible documentation, productionization spec, report, and policy/community brief; present to CUSP and civic stakeholders.	Engine + pilot + documentation + handoff

> **The plan is sized to CUSP's standard three-student team (~225 committed hours per semester). Students build the engine and a pilot — not a production system. If CUSP reads this as two projects, the engine/product seam is the natural split.**

EXPECTED DELIVERABLES

07 SEVEN COMMITTED. THE REST CONDITIONAL.

COMMITTED

- **1 · Provenance Evidence Pipeline.** Documented, version-controlled ingestion, entity resolution, evidence classification, and temporal provenance processing.
- **2 · Temporal Knowledge Graph & Provenance Index.** Auditable people–place–event relationships and a transparent parcel-level evidence score.
- **3 · Validation Result.** Corridor-level held-out testing, the connectivity-versus-proximity analysis, robustness testing, and a documented decision on whether a monetary interpretation may be displayed.
- **4 · Working Greenmount/Waverly Pilot.** The three-part address experience. The first two panels — documented history and observed value/provenance evidence — are committed; the third shows applicable programs and evidence-backed actions in all cases, with a projected post-improvement value appearing only if the conditional scenario model meets its data-quality and validation thresholds.
- **5 · Model Card & Public-Use Guardrails.** Intended uses, prohibited uses, uncertainty, known bias, data gaps, and failure conditions.
- **6 · API/Data Contract & Productionization Specification.** The documentation BNBD/Niceworks uses to deploy and maintain the engine.
- **7 · Reproducible Dataset, Code, Report & Community Brief.** Subject to restricted-data and licensing limitations.

CONDITIONAL

- CONDITIONAL National Register causal analysis · equity heterogeneity extension · improvement-scenario model · expanded program knowledge base · automated source connectors · one creative prototype · open-data memorandum.

ETHICS, OPERATIONS & RISKS

08 PUBLIC DUTIES, NAMED RISKS

- **Human subjects.** The committed research core relies on public administrative, property, and archival records and is anticipated not to constitute human-subjects research; the team will confirm the appropriate NYU HRPP determination before analysis begins. The existing tools' oral-history functionality sits outside the committed research dataset unless separately reviewed.
- **Restricted data.** Any ZTRAX use occurs under ICPSR's agreement inside CUSP's secure Research Computing Facility; no restricted microdata leaves the enclave; only aggregate or derived outputs ship.
- **Intellectual property.** The sponsor prefers an MIT-licensed open-source release of the capstone code and index specification — students retain ownership and full credit; BNBD/Niceworks productionizes from the open release — subject to the CUSP client-team agreement, restricted-data requirements, and applicable NYU approvals, and memorialized before work begins.
- **A public tool has public duties.** Confidence intervals and traceable sources on every output; unverified history labeled, never asserted; no appraisal language, no guaranteed-return language, no automatic eligibility determinations; resident-and-owner usability and anti-displacement framing; model-card disclosure; and the ability to suppress a monetary estimate whenever validation is insufficient.
- **Independence from government.** The project is independent civic research and tooling: it is not commissioned by, performed for, or delivered to the City of Baltimore or any government entity. Any future public-sector adoption would follow that entity's own review and procurement processes.
- **Maintainability.** Students deliver the engine + pilot + specification; **BNBD / Niceworks productionizes and maintains** the public tool against refreshed public data. No student-maintained production system.

RISKS & MITIGATIONS

08 NAMED, NOT HIDDEN

RISK	MITIGATION
Index fails its validation threshold	The pilot ships the documented evidence layer without a monetary estimate — pre-committed as the honest outcome, and the tool remains useful.
Circularity / temporal leakage	Three-stage design: pre-specified score, outcome-blinded coding, held-out validation; time-stamped claims reconstructed as of outcome dates.
Archival abundance mistaken for historical significance	Evidence availability is separated from significance; every score carries a source-coverage/completeness measure; low documented evidence is never interpreted as "no history"; archival silence is identified explicitly; reviewed community-supplied evidence is admissible without treating unsupported claims as verified.
Sponsor development interests create an actual or perceived conflict	Methodology and model card are public; relevant sponsor property interests are disclosed; the sponsor cannot override evidence classifications or model results; faculty and student analytical independence is preserved; bulk "undervalued-property" rankings are prohibited; public outputs are designed for parcel understanding, not acquisition targeting.
Restricted-data access (ZTRAX) delayed or insufficient	Go/no-go decision in early spring; corridor core never depends on it; external geography narrows to available data.
No clean "eligible-but-not-listed" control	Federal DOE file rejected as unsuitable; if the external analysis proceeds, not-yet-listed properties serve as controls in a staggered design.
Unmaintainable prototype	Students hand off engine, pilot, model card, and spec; the sponsor owns production. The graded artifact is the data science, not the app.
Scope creep	Committed / conditional / later-phase tiers throughout; conditional work is cut first; at most one prototype module.
Public over-claiming or harm	Three-value separation (observed / evidence / scenario), guardrails above, and suppression rules baked into the pilot.

WHAT THE SPONSOR PROVIDES

09 A RUNNING HEAD START

- **Two live tools** (the 22-source Property Dossier and the Improve program-matcher) — proven interface and live-data plumbing the engine plugs into, so students build intelligence, not infrastructure.
- **The labeled ground truth:** the curated corridor knowledge base (~354 properties, ~2,600 incidents, ~150 registered intellectual-property records) plus the 139 digitized Waverly designation claims.
- **The research roadmap:** the *Maryland & Baltimore Property Research Index* — a curated, battle-tested catalog of 100+ property-research sources with access paths documented.
- **Funded, state-backed documentation:** the active MHT FY2026 Waverly Main Street Study (18 months, assigned project monitor) producing per-property white papers during the capstone.
- **Implementation capacity:** the sponsor's active corridor work — documented in its Opportunity Zone 2.0 nomination case — provides a credible pathway for field-testing and long-term stewardship, subject to the proposal's public-use, transparency, and conflict-of-interest safeguards (§08).
- **Productionization and maintenance:** BNBD/Niceworks will take the engine to production and maintain it, while providing domain guidance and regular sponsor check-ins of approximately 2–5 hours per week, with heavier engagement during project initiation.

→ POSITIONING: PHASE 1 OF A LARGER PROGRAM

The sponsor's *Empirical Economics of Provenance* program spans provenance measurement, recognition capitalization, narrative connectivity, historical devaluation, improvement uplift, certification and recognition experiments, diffusion across connected places, activation-and-protection design, and public property-intelligence infrastructure. **The capstone is not that program. It builds the measurement infrastructure that makes the program possible** — and everything a city could put in residents' hands runs on this layer being built correctly first.

> **One defensible evidence engine. One validated corridor pilot. A disciplined pathway to external testing. A foundation for everything that follows.**

SOURCES & REFERENCES

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NPS — National Register Data Downloads (national-register-listed.xlsx; federal-DOEs.xlsx). nps.gov/subjects/nationalregister/data-downloads.htm · NPS IRMA DataStore Reference 2210280.

ICPSR — Zillow Transaction and Assessment Database (ZTRAX), 1940–2020 (Study 39652). icpsr.umich.edu/web/ICPSR/studies/39652 (restricted-use).

Mapping Inequality: Redlining in New Deal America (HOLC maps), University of Richmond. dsl.richmond.edu/panorama/redlining

U.S. Census Bureau — American Community Survey 2020–2024 5-Year Estimates (tables B19113, B17001) and TIGER/Line 2020 census-tract geometry. data.census.gov

Open Baltimore — parcel, permit, and vacancy datasets (data.baltimorecity.gov) · Maryland State Department of Assessments & Taxation, Real Property data.

Waverly Main Street Historic District — NRHP ref. 13001020, listed December 31, 2013; 168 resources, 143 contributing buildings.

City of Baltimore — vacant-property reduction framework (≈37,500 properties; ≈$5B private-investment target), per Mayor's Office and Greater Baltimore Committee public materials, 2023–2026.

Sponsor-prepared: Greenmount Corridor Opportunity Zone 2.0 Nomination Case (REV02, July 2026) — corridor parcel, ownership, vacancy (occupancy census), and market-study figures, disclosed as sponsor-prepared throughout.

NYU CUSP — Capstone Projects Sponsor Guide. engineering.nyu.edu/research/centers/cusp/engagement/capstone-projects-sponsor-guide